

Fractions — Diagnostic Assessment

Ghanaian upper primary (B.S. 4-6), generated against a 10-node skill graph.

Class:

Date:

Answer every question. For each item, **circle the letter** of the best answer. Working may be shown on the back of the page.

- 1. What is 6×7 ?**

B. 40

D. 48

2. Kofi has 15 baskets of mangoes. Each basket holds 30 mangoes. How many mangoes does he have in total?

B. 150

D. 4500

- 3.** Ama wants to share 17 oranges equally among 4 friends. How many oranges does each friend get, and how many are left over?

B. 4 each, 0 left over

D. 5 each, 3 left over

4. What is $3 \div 6$?

B. 0.5

D. Cannot be done

- 5.** A rectangle is divided into 5 equal parts. 2 of those parts are shaded. What fraction of the rectangle is shaded?

B. 2/5

D. 5/2

- 6.** Two friends each cut a kelewele plate. Kofi cuts his into 4 equal pieces and shades 1. Ama cuts hers into 4 pieces of different sizes and shades 1. Whose shaded part is exactly $\frac{1}{4}$ of the whole?

B. Kofi only

D. Neither

7. Which fraction is equivalent to $\frac{2}{3}$?

B. 3/4

D. 5/6

8. Fill in the missing number: $\frac{3}{5} = \frac{?}{15}$

A. 8

B. 9

C. 10

D. 13

9. Simplify $\frac{6}{9}$ to its lowest terms.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{3}{4}$

D. $\frac{6}{9}$

10. Simplify $\frac{12}{18}$.

A. $\frac{1}{2}$

B. $\frac{2}{3}$

C. $\frac{4}{6}$

D. $\frac{6}{9}$

11. Convert $2\frac{1}{3}$ to an improper fraction.

A. $\frac{2}{3}$

B. $\frac{3}{3}$

C. $\frac{6}{3}$

D. $\frac{7}{3}$

12. Ama bought $3\frac{3}{4}$ kg of yam at the market. Express this as an improper fraction in kg.

A. $\frac{7}{4}$

B. $\frac{12}{4}$

C. $\frac{13}{4}$

D. $\frac{15}{4}$

13. Convert $\frac{11}{4}$ to a mixed number.

A. $2\frac{1}{4}$

B. $2\frac{3}{4}$

C. $3\frac{1}{4}$

D. $\frac{11}{4}$

14. Akua walked $\frac{17}{5}$ km from her home to the market. Write this distance as a mixed number.

A. $2\frac{2}{5}$

B. $3\frac{2}{5}$

C. $3\frac{3}{5}$

D. $5\frac{3}{5}$

15. Compute $\frac{2}{3} \times \frac{4}{5}$.

A. $\frac{6}{8}$

B. $\frac{8}{8}$

C. $\frac{8}{15}$

D. $\frac{10}{12}$

16. Yaa has $\frac{3}{4}$ kg of rice. She uses $\frac{2}{5}$ of it to cook jollof. How much rice did she use, in kg?

A. $\frac{5}{9}$

B. $\frac{5}{20}$

C. $\frac{6}{20}$

D. $\frac{5}{4}$

17. Compute $\frac{3}{4} \div \frac{1}{2}$.

A. $\frac{1}{2}$

B. $\frac{3}{8}$

C. $\frac{6}{4}$

D. $\frac{8}{3}$

18. Kwame has $\frac{6}{8}$ of a tray of meat pies. He divides his portion into pieces, each one $\frac{1}{4}$ of a full tray. How many pieces does he get?

A. $\frac{1}{3}$

B. 3

C. $\frac{6}{32}$

D. $\frac{32}{6}$

19. Compute $\frac{1}{2} \times \frac{2}{3} \div \frac{1}{4}$.

A. $\frac{1}{3}$

B. $\frac{1}{12}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

20. Compute $(\frac{2}{3} \times \frac{3}{4}) \div \frac{1}{2}$.

A. 1

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. $\frac{1}{8}$

21. Compute $1\frac{3}{4} \times \frac{2}{3}$.

A. $\frac{2}{3}$

B. $\frac{7}{6}$

C. $\frac{5}{7}$

D. $\frac{21}{12}$

22. Esi wrote $\frac{12}{18}$ on her exercise book. Express this as a fraction with denominator 3.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{4}{3}$

D. $\frac{6}{3}$

23. Compute $\frac{4}{6} \times \frac{3}{8}$ and give your answer in lowest terms.

A. $\frac{1}{4}$

B. $\frac{1}{3}$

C. $\frac{7}{14}$

D. $\frac{12}{48}$

24. Compute $1\frac{1}{2} \times \frac{2}{3} \div \frac{1}{4}$.

A. $\frac{1}{4}$

B. 1

C. 4

D. $\frac{3}{8}$

25. Compute $1\frac{1}{2} \times 2\frac{1}{2}$ and write the answer as a mixed number.

A. $2\frac{1}{2}$

B. $3\frac{3}{4}$

C. $\frac{5}{4}$

D. $\frac{15}{4}$

26. Compute $2\frac{1}{2} \div 1\frac{1}{4} \times \frac{2}{3}$.

A. $\frac{1}{3}$

B. 2

C. $\frac{4}{3}$

D. $\frac{25}{12}$